

Project Requirement Document

DAY 1 — Understand the Problem

Project: Automatic Skill Extraction from Job Descriptions

Task Overview

The intern must understand the following concepts before moving forward with the project:

- What is a job description?
- What is a skill?
- Why automatic skill extraction is useful?
- Difference between keyword extraction, entity extraction, classification, and information extraction.
- Define project input and output.

1. What is a Job Description?

A Job Description (JD) is the description of a job role that contains information about the role and the skills or experience required for that position. In this project, the job description is the main text from which useful information such as the role and skills will be extracted.

2. What is a Skill?

A skill is a knowledge area, technology, tool, or capability required for a job role. In the project example, Python, Pandas, NumPy, Scikit-learn, SQL, and AWS are the skills mentioned for a Data Scientist role.

3. Why Automatic Skill Extraction is Useful?

Automatic skill extraction is useful because job descriptions can contain many pieces of information. Automatically identifying the required skills can make the process of finding relevant skills from job descriptions easier and faster.

4. Difference Between Key Concepts

Keyword Extraction: Identifying important words or phrases from a piece of text.

Entity Extraction: Identifying specific entities or meaningful items in text, such as a job role or a skill.

Classification: Assigning a text or item to a predefined category.

Information Extraction: Extracting useful information from text and representing it in a structured form.

5. Project Input and Output

Input: A Job Description in text form.

Example Input:

We require a Data Scientist with Python, Pandas, NumPy, Scikit-learn, SQL and AWS experience.

Expected Output: The system should identify the role and skills from the job description in a structured format.

Example Output:

```
{  
  "role": "Data Scientist",  
  "skills": [  
    "Python",  
    "Pandas",  
    "NumPy",  
    "Scikit-learn",  
    "SQL",  
    "AWS"  
  ]  
}
```

6. Project Document Requirements

The project document should contain the following sections:

- Problem Statement
- Business Objective
- Technical Objective
- Input
- Expected Output
- Technologies
- Success Criteria

7. Problem Statement

Job descriptions contain useful information such as job roles and required skills. The project focuses on understanding how this information can be identified automatically from job description text.

8. Business Objective

The business objective is to make the process of identifying required skills from job descriptions easier and more efficient through automatic skill extraction.

9. Technical Objective

The technical objective is to use the listed technologies and NLP-related approaches to process job descriptions and extract relevant information such as the job role and skills.

10. Input

The project input is a job description in text format.

11. Expected Output

The expected output is structured information containing the job role and the skills extracted from the job description.

12. Technologies

The technologies listed in the Day 1 task are:

- Python
- Jupyter Notebook
- Pandas
- NumPy
- NLTK
- spaCy
- Scikit-learn
- Transformers
- Matplotlib
- Seaborn
- Power BI/Tableau
- Flask/FastAPI

13. Success Criteria

The project should be able to take a job description as input and produce structured information identifying the role and relevant skills from that job description.